

Design and Implementation of the “Elder Care Without Worries” Integrated Smart Elderly Care Platform

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Abstract

The “Elder Care without Worries” Integrated Smart Elderly Care Service Platform is designed with the principles of professionalism, digitalization, intelligence and humanization, focusing on “Elderly Health Companionship”. This platform integrates cutting-edge smart technologies with meticulous care services to address the aging population trend in China. It employs real-time database analysis for elderly health monitoring and companionship. The platform features functions such as health management, online virtual moments, cloud communication, global network information, dynamic guidance, and price comparison. The construction of this smart elderly care platform not only enhances the quality of life for the elderly but also promotes the popularization and development of smart elderly care services, contributing to a more humanized social elderly care system.

CCS Concepts

• **Software and its engineering** → Software creation and management; Designing software.

Keywords

smart elderly care, elderly, integrated services, health management

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1 INTRODUCTION

With the increasing number of elderly people, the trend of population aging has intensified, bringing the spiritual and cultural life of the elderly into focus and attracting wide attention from society^[2]. With the country promoting the policy of smart elderly care services, it is evident that there is currently a lack of simple and convenient smart elderly care services for the elderly. It is necessary to utilize modern technology to provide safe, healthy, and convenient smart services for the elderly^[3]. Therefore, according to the needs of the elderly for smart old-age care, this paper devotes itself to combining the latest technical level with the concept of smart old-age care^[5], and builds an intelligent old-age care system including news, chat, class, shopping, health inspection and navigation, etc., which can meet the needs of the elderly for smart old-age care services from all aspects. The system aims to enable the elderly to enjoy high-quality smart elderly care services easily and conveniently, thereby improving their happiness index^[6].

In addition, the article has the following innovative points.

(1) Innovation of ideas

The project upholds the concept of humanistic innovation. It centers on the needs and well-being of the elderly and puts the well-being of the elderly first.

Innovation with sustainable development concept. Pay attention to long-term sustainable service model for the elderly.

(2) Innovation of functional design

Each plate of the project contains the power of innovation, each plate is like a shining star, unique, dazzling.

The online shopping sector leads the consumption pattern of the elderly with its innovative model.

With its advanced technology and innovative methods, the health detection sector provides comprehensive health management services for the elderly.

The elderly care course sector is another innovative sector.

The online chat board breaks the loneliness and isolation of the elderly with its innovative way of socializing.

The news browsing section is another innovative feature.

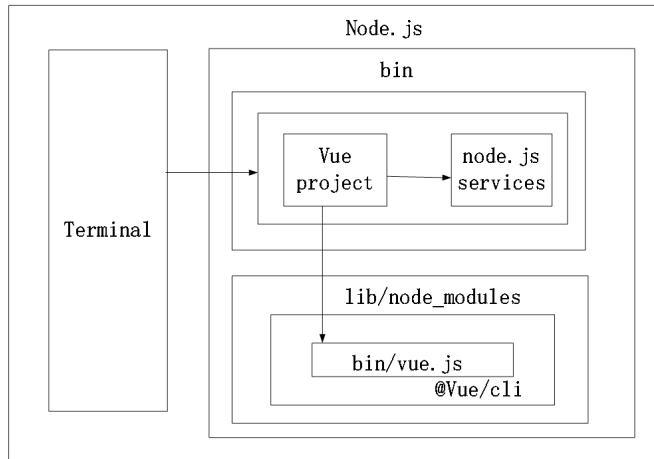


Figure 1: System structure frame diagram.

2 SYSTEM REQUIREMENTS AND ARCHITECTURE DESIGN

In order to meet the needs of intelligent old-age care, the design and implementation of the intelligent old-age care system follow the requirements of open humanized design and cross-platform ability, and applies the latest technology to meet the needs of the elderly. The system adopts advanced technology and design concept to adapt to different devices through responsive design^[8], and adopts artificial intelligence technology to provide personalized recommendation with functions such as identification and navigation to ensure that the elderly can enjoy the convenience brought by the latest scientific and technological achievements. The system can digitally model the layout of the elderly's homes^[9], which is convenient for the children of the elderly to view indoor data, the health parameters of the elderly, and the home situation of the elderly through the system. The system can collect the health status of the elderly and give corresponding medical suggestions based on the physical condition information of the elderly. It can also predict the occurrence of major diseases so that the elderly can seek medical treatment in time^[1]. The frame diagram of the system structure is shown in Figure 1.

The system architecture is composed of a client layer, Web application layer, system layer, and terminal layer. Node.js is used as the scaffold, and the JavaScript language is adopted^[7]. This system uses Node.js based on the Chrome V8 engine as JavaScript running on the server side, allowing people to use JavaScript to write server-side applications. With the application of Node.js, JavaScript not only can be used to build interactive front-end applications, but also to build high-performance and extensible back-end services. At the same time, it also solves the problem of weak portability of the Web. The system adopts Vue-cli scaffolding on the project structure and pre-configured construction tools. Its flexibility and rich tool components greatly reduce the time and energy consumption of personnel in project setting and configuration^[7]. Through reasonable system architecture design, the "Worry-free for the Elderly" smart pension integration platform is an extensible, flexible, and

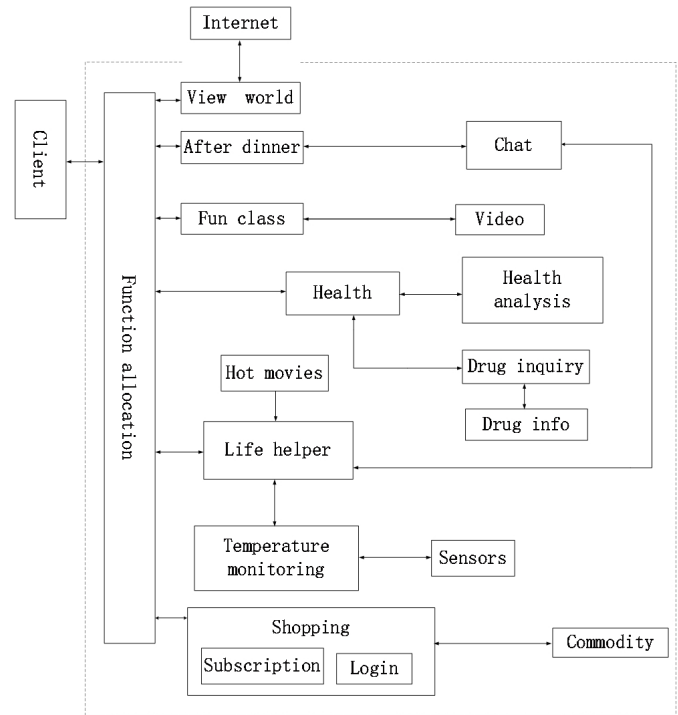


Figure 2: System service flow chart.

stable system providing efficient, safe, and credible smart pension services for the elderly.

3 INTEGRATED PLATFORM DESIGN

3.1 System Function Design

In order to solve the problem that many systems related to aged care have single functions, too much irrelevant information^[4], and insufficient intelligence, this system carefully constructs an all-round service system for the aged care based on interdisciplinary theoretical basis and practice "Worry-free for the elderly" means that the elderly will be used as a service group to provide a professional, digital, intelligent and humanistic health service system, and fully advocate a scientific, advanced and convenient new concept of old-age care. The system aims to respond to the professional needs of the elderly in health management, education, social activities, and daily shopping, and constructs a highly integrated online service platform. The service flow is shown in Figure 2.

With its holistic and inclusive design concept, the system covers comprehensive service functions (as shown in Figure 2), including "sitting in the world," "having fun after dinner," "classroom," "shopping in peace," "shopping in colorful ways," "life helpers," and other functions, and is committed to providing a one-stop network experience for elderly users. The interface function settings are shown in Figure 3.

In the aspect of health management, the Ping An'an module mainly collects human body sign parameters. This module can provide scientific and reasonable suggestions. In terms of education and entertainment, the old-age course module tailors a series of

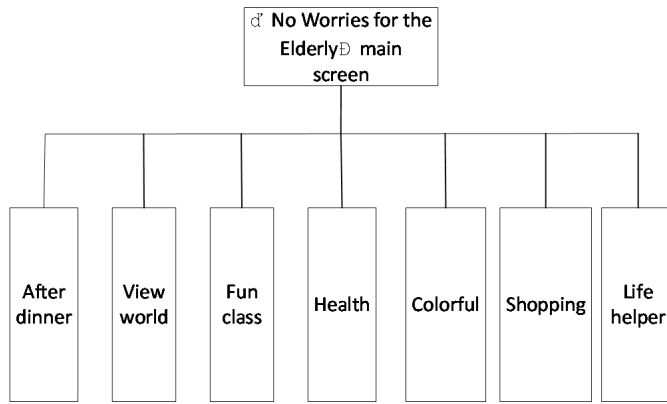


Figure 3: "Worry-free for the Elderly" Smart Pension Integration Platform Interface.

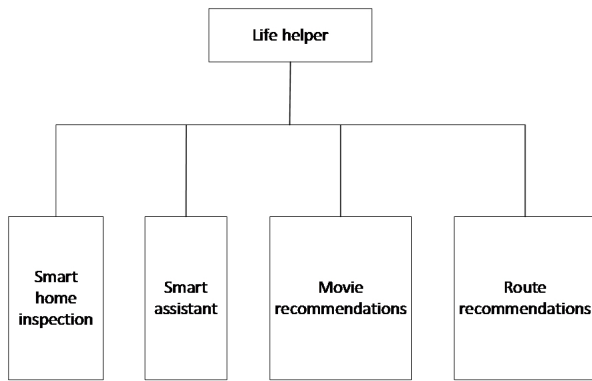


Figure 4: "Life Helper" service interface.

course contents for elderly users. The after-dinner module of the system creates a new type of communication space for users to connect with family, friends, and community members through the virtual address book and contact list. The online shopping module of the system improves the shopping convenience and pleasure of elderly users. Users can easily browse, purchase, and settle goods. Every step is carefully designed to provide an easy-to-use and safe shopping experience. In the colorful module, according to the preferences of the elderly, the system recommends many web pages with high click rates related to the elderly.

As a comprehensive platform, the Life Helper module includes a series of practical functions such as intelligent assistant, intelligent home monitoring, movie recommendation, and route inquiry (as shown in Figure 4). The intelligent assistant function allows the elderly to alleviate negative emotions brought by loneliness by chatting with their children. The smart home detection function can detect the situation at home and facilitate children in understanding the situation of the elderly. The movie recommendation function recommends various types of movies to the elderly and can link to various film and television platforms. The route search function is convenient for the elderly to plan their subway routes without

leaving home. These designs can save a lot of energy for the elderly in aspects of traveling, watching movies, and chatting.

3.2 Implementation of key technologies of the system

This system adopts an intelligent recommendation algorithm in the specific recipe function of "PingPingAnAn", which is based on the idea of hash function. The data input by the customer is stored in the memory space pointed to by the structure pointer, and the corresponding relationship between the sign code and the recommended food is stored through the hash table. When BMI is less than 18.5, the length of the sign code is 4; when BMI is between 18.5 and 23.9, the length of the sign code is 6; when BMI is greater than or equal to 24, the length of the sign code is 8. In the BMI formula, w is considered body weight, and h is considered height. The calculation formula of BMI is as follows:

$$BMI = \frac{W}{H^2} \quad (1)$$

After dividing the first priority BMI, the system will judge the blood sugar, blood lipid, and blood pressure respectively and encode the corresponding signs. Among them, the sign codes for blood lipid, blood glucose, and blood pressure are 000110, and the sign codes for high normal and low degree are also 000110 (as shown in Figure 5). According to the data input by the user, the system will return a physical sign code to the database through the judgment of the above algorithm.

After receiving the data from the front end of the database, the mapping relationship between food and sign code is found based on this algorithm, and two database relationship tables are designed, which are called food table and sign code table respectively. The food table contains various attributes of food, such as food number, name, calorie, fat fiber and label code^[7]. The sign coding table serves as a mapping and interpretation function of data, including a sign coding field and a description field. First, select statement is used to retrieve the data in the sign code interpretation table to obtain the corresponding sign status.

The combination of select and where query statement can return the sign code interpretation, which is similar to the situation of low body weight and high blood lipid. Such a query operation can help the system understand the user's health status and provide the basis for subsequent health recommendations and food recommendations. Secondly, when recommending food, the system adds the qualified food to the recommended list by traversing the food table. If the sign code set of the food table is a subset of the user's sign code, the system selects the specific sign code and its associated food information by combining the food table and the sign code table with the SQL statement composed of keywords such as SELECT JOIN and WHERE^[10], so as to realize the function of recommending the corresponding food according to the user's sign. The database relational table design is shown in the Table 1 and Table 2.

The system utilizes the ChatGPT large model API to create an intelligent assistant page, significantly enhancing the system's intelligence and facilitating daily life consultations for the elderly. ChatGPT is a language model based on the GPT (Generative Pre-trained Transformer) architecture. Its core functionality involves

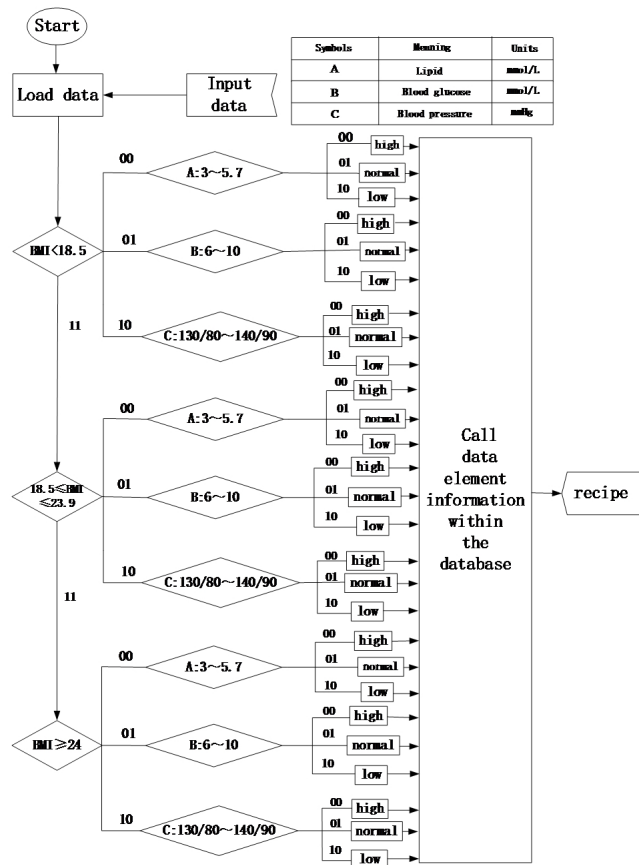


Figure 5: Flow chart of intelligent recipe recommendation algorithm.

Table 1: Database food relationship diagram.

Cross-reference List	
Column Name	Description
foodID	Food number
foodName	Food name
calories	Calorie
fat	Fat
fiber	Fiber
category	Label coding

pre-training on a vast amount of textual data, enabling it to understand context and generate coherent natural language text. The API (Application Programming Interface) operates by defining a set of rules and protocols that allow different software systems to communicate and interact with each other. The model, with its numerous parameters and hierarchical structure, simulates the complexity and diversity of human language. ChatGPT API, provided by OpenAI, allows developers to interact with the ChatGPT large model through HTTP requests. The API is designed to simplify natural language processing tasks, enabling users to easily

Table 2: The relationship table for database sign coding.

Sign Coding Table	
Sign Coding	Description
0000, 110000, 11110000	High Blood Lipids in Underweight, Normal Weight, and Overweight Individuals
0001, 110001, 11110001	Normal Blood Lipids in Underweight, Normal Weight, and Overweight Individuals
0010, 110010, 11110010	Low Blood Lipids in Underweight, Normal Weight, and Overweight Individuals
0100, 110100, 11110100	High Blood Glucose in Underweight, Normal Weight, and Overweight Individuals
0101, 110101, 11110101	Normal Blood Glucose in Underweight, Normal Weight, and Overweight Individuals
0110, 110110, 11110110	Low Blood Glucose in Underweight, Normal Weight, and Overweight Individuals
1000, 111000, 11111000	High Blood Pressure in Underweight, Normal Weight, and Overweight Individuals
1001, 111001, 11111001	Normal Blood Pressure in Underweight, Normal Weight, and Overweight Individuals
1010, 111010, 11111010	Low Blood Pressure in Underweight, Normal Weight, and Overweight Individuals

integrate powerful language generation capabilities into various applications. The core idea of the API is stateless communication between the client and server, where each request contains all the necessary information, ensuring high availability and scalability of the system. The use of the HTTP protocol not only simplifies request and response handling but also ensures good compatibility with existing network infrastructure. Below is the code and technical description of the ChatGPT large model API call:

In this system, the code first uses the requests library to send HTTP requests and the json library to handle JSON data. These libraries can be installed via the following command: `pip install requests`. The code defines the API URL and API key, replacing the placeholder 'your_api_key_here' with the actual API key, and specifies the API URL with `api_endpoint`. The request headers contain authentication information (i.e., the API key) and the content type (JSON), which are crucial for ensuring the server can correctly authenticate and parse the request^[5]. The Content-Type is set to `application/json`, and the Authorization field includes the Bearer token. On this basis, the request data is constructed, defining the model name (e.g., `gpt-4`), the list of dialogue messages (messages), the temperature parameter (which controls the randomness of the generated text), and the maximum number of tokens (which limits the length of the response). The call to the API is completed by using `requests.post` to send the request to the API endpoint and pass the request header and request body data. API responses typically contain status codes and error messages that indicate the result of the request and potential problems.

In the functional technology of drug query, the data resources in this system are stored in the background database. Connection technology uses mysql module interface provided by javascript to establish connection with database and send query request^[11]. In the concrete implementation, the front-end application sends

Algorithm 1 Interface invocation Algorithm.

```

import requests
import json
# Set API key and endpoint
api_key = 'your_api_key_here'
api_endpoint = 'https://api.openai.com/v1/chat/completions'
# Construct request headers
headers = {
    'Content-Type': 'application/json',
    'Authorization': f'Bearer {api_key}',
}
# Construct request data
data = {
    'model': 'gpt-4',
    'messages': [
        {'role': 'system', 'content': 'You are a helpful assistant.'},
        {'role': 'user', 'content': 'Can you explain how to use the
ChatGPT API?'},
    ],
    'temperature': 0.7,
    'max_tokens': 150,
}
# Send request
response = requests.post(api_endpoint, headers=headers,
json=data)
# Handle response
if response.status_code == 200:
    result = response.json()
    print("Response from ChatGPT:")
    print(result['choices'][0]['message']['content'])
else:
    print(f"Error: {response.status_code}, {response.text}")
import requests
import json

```

query condition to Node.js server through HTTP request After the Node.js server receives the request, it uses the mysql module of javascript to establish the connection and forward the query The background database is established by using the relational database mysql. In the establishment of mysql, we strictly follow the three normal rules^[7]. The relationship between tables is generated by foreign keys, and the concepts of primary key and unique key are adopted in the table to ensure the integrity and standardization of

data For example, the database relational table of drug information in the drug query subsystem is designed as shown in Table 3 below.

The drug number field in the table is of integer type and self-increasing primary key, which ensures the uniqueness of each record The types of English name and other name fields of drugs are variable length strings, which saves memory space However, the attributes of pharmacological information, such as indications, dosage, usage and adverse reactions, which involve large text, are stored by text type in the database^{[10][9]}. The advantage of using text type to store data is that modern relational database optimizes its query and supports full-text retrieval function, which can efficiently perform complex text search and matching operations.

4 SYSTEM TESTING

Several testers were divided into five groups, and each group input different data of height^[12], weight, blood pressure, blood sugar and blood lipid to the healthy recipe recommendation interface of this system The test contents include: the first group inputs the index value in the normal range to observe whether the recommendation of the program is reasonable in the healthy state; In the second group, high blood glucose levels were input to verify whether the program could correctly provide dietary suggestions suitable for patients with diabetes or hyperglycemia; The third group entered high blood pressure values to test whether the program can identify the needs of hypertensive patients and provide a low-salt and low-fat diet^[13]; The fourth group entered high blood lipid values to check whether the program would recommend a diet beneficial to hyperlipidemia patients; In the fifth group, the low body weight was input to observe whether the program could give a diet with high calorie and high nutrition suitable for those with low weight Each group of experiments should be repeated two or three times throughout the testtext, are stored by text type in the database. To ensure the stability of the program and the consistency of the recommended results Record the recommended breakfast, lunch, dinner and feedback of the program after each test to ensure that the performance under different health conditions meets the expected test results as shown in Table 4.

5 CONCLUDING REMARKS

With the emergence of the “Worry-Free” smart elderly service system for the elderly population, it can demonstrate significant practical value. With the goal of improving the lives of the elderly and promoting the popularization of smart old-age services^[14], the system integrates a number of key functions including news

Table 3: Database drug relationship table.

Drug Information Table			
Column Names	Description	Column Names	Description
Medication ID	Drug Code	usage	Dosage and Administration
alias	Drug Alias	Adverse Reactions	Adverse Reactions
foreign Name	English Name of Drug	Contrain dications	Contraindications
pharmacology	Pharmacological Information	precautions	Precautions
indications	Indications	Drug Interactions	Drug Interaction Information

Table 4: Table of system test results.

Test ID	Height	Weight	Is the catering recommendation normal
1	170	65	normal
2	160	55	normal
3	180	75	normal
4	165	60	normal
5	175	70	normal

browsing, online social courses, learning shopping convenience, health monitoring and navigation, etc. We fully consider the operation habits and special needs of the elderly, making the system interface simple and easy to understand and convenient to operate, and adopting advanced technology to ensure the stability, safety and high efficiency of the system. In the future, the "Worry-Free Elderly" system will further improve the "Worry-Free Elderly" smart elderly service system and build a more perfect smart elderly ecosystem to provide more comprehensive, convenient, and humanized services for the elderly. This paper focuses on the design and implementation of intelligent pension service system, and discusses the intelligent pension system based on JavaScriptWeb and database. The system uses Node.js as the scaffold by Vue-cli and Javascript language and combines with a variety of modern technical means to build an extensible, flexible and stable integrated platform. It aims to provide a safe, healthy and convenient intelligent service for the elderly. Using intelligent assistant to call big language model to provide convenience and care for the life of the elderly. The platform refines the physiological indicators of users and gives reasonable dietary suggestions according to the intelligent recipe recommendation algorithm, and provides an intelligent drug search function to ensure the health and rationality of the elderly. This system adopts cutting-edge framework design and combines various modern technologies to protect the convenience, health, and happiness of the elderly.

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